

PHISHING AWARENESS CAMPAIGN

Security Analysis Report

Hadiya Zuberi

Instructor: Mahnoor Gillani

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1. Introduction

Phishing is one of the most prevalent and effective forms of social engineering used by cybercriminals to steal credentials, gain unauthorized access to systems, or deploy malware. Unlike technical exploits, phishing attacks target human psychology, exploiting emotions such as fear, urgency, and trust to manipulate victims into taking harmful actions.

According to global cybersecurity statistics, phishing remains responsible for over 80% of reported security incidents in organizations worldwide. Despite advances in technical defenses, human error continues to be the most exploited vulnerability in any security chain.

This report documents a controlled phishing simulation exercise conducted as part of a university cybersecurity course. The exercise was designed to evaluate how susceptible individuals are to a realistic phishing attempt, identify the psychological triggers that lead to compliance, and develop actionable recommendations for improving phishing awareness.

2. Objectives

The primary objectives of this phishing awareness campaign were:

- To design a realistic phishing email that mimics a legitimate institutional communication.
- To simulate an attack scenario in a safe, ethical, and controlled environment.
- To measure the susceptibility rate among a sample group of university students.
- To identify the specific psychological triggers that contributed to the attack's success.
- To provide targeted recommendations for improving phishing resilience.

3. Ethical Considerations & Consent

This exercise was conducted strictly within ethical boundaries. All participants were informed in advance that they would receive a simulated phishing attempt within a defined timeframe.

The following ethical principles were observed throughout the exercise:

- No real credentials, passwords, or sensitive personal data were collected at any point.
- The Google Form used as a landing page did not store or transmit any login information.

- All participants were fully debriefed immediately after submitting the form via an on-screen awareness banner.
- Participant identities and response data are kept confidential and used solely for academic analysis.

4. Methodology

4.1 Participant Selection

A group of 6 university students from Air University Aerospace and Aviation Campus, Kamra were selected as participants. All participants were peers who provided informed consent prior to the exercise. The sample was intentionally kept small given the academic scope of this project.

4.2 Tools Used

Tool	Purpose
Gmail	Sending the phishing email from a spoofed name account
Google Forms	Simulated credential entry landing page
Google Sheets	Automatic logging of form submissions and timestamps
Ctrl+K Hyperlink	Creating deceptive display URL vs real URL mismatch

4.3 Data Collection

Success was measured by tracking form submissions via Google Sheets. Each submission automatically recorded the participant's entered name/ID and timestamp. Two metrics were tracked:

- Link clicked and form submitted (primary success metric)
- No action taken, email ignored or identified as suspicious (awareness metric)

4.4 Timeline

Phase	Activity
Phase 1	Participant consent obtained
Phase 2	Phishing email designed and reviewed
Phase 3	Email sent to all 6 participants
Phase 4	Responses monitored via Google Sheets
Phase 5	Participants debriefed and results analyzed
Phase 6	Report compiled

5. Attack Design

5.1 Scenario

The attack scenario was designed to impersonate the IT Support Team of Air University, informing students that a mandatory password reset was required as part of a system upgrade to a new Student Portal. This scenario was selected because it is highly believable in a university context, especially around system migration periods.

5.2 Phishing Email

The final phishing email incorporated the following elements to maximize believability:

Element	Content	Psychological Purpose
Sender Display Name	AiruniversityKamra IT_Department	Impersonates official IT team
Subject Line	Password Reset Required — Action Needed by March 3rd	Creates urgency and specificity
Deadline	March 3rd, 2026	Artificial time pressure
Consequence	Account temporarily suspended	Fear-based motivation
Link Display Text	portal.airuniversity- portal.edu.pk/reset	Appears legitimate
Actual Link	Google Forms URL	Real destination hidden
Signature	Full name, title, helpdesk email, phone	Establishes authority
Disregard Line	If already reset, disregard	Reduces suspicion

Screenshot 1: Final Phishing Email as received by participants

Password Reset Required — Action Needed by March 3rd

External

Inbox x



AiruniversityKamra IT_Department <airuniversitykamraitdepartment@gmail.com>

to hassanrabya90, me, 245011@aack.au.edu.pk, 245047@aack.au.edu.pk, 245051@aack.au.edu.pk, 245053@aack.au.edu.pk ▼

Dear student,

As part of our system upgrade to the new Student Portal, all students are required to verify and reset their account passwords.

You must complete this before March 3rd, 2026, or your account will be temporarily suspended.

Please reset your password using the secure link below:

<portal.airuniversity-portal.edu.pk/reset>

If you have already reset your password, please disregard this email.

Regards,

Ali Hassan

IT Support Team

Air University Aerospace and Aviation Campus, Kamra

helpdesk@airuniversitykamra.edu.pk | 051-517263

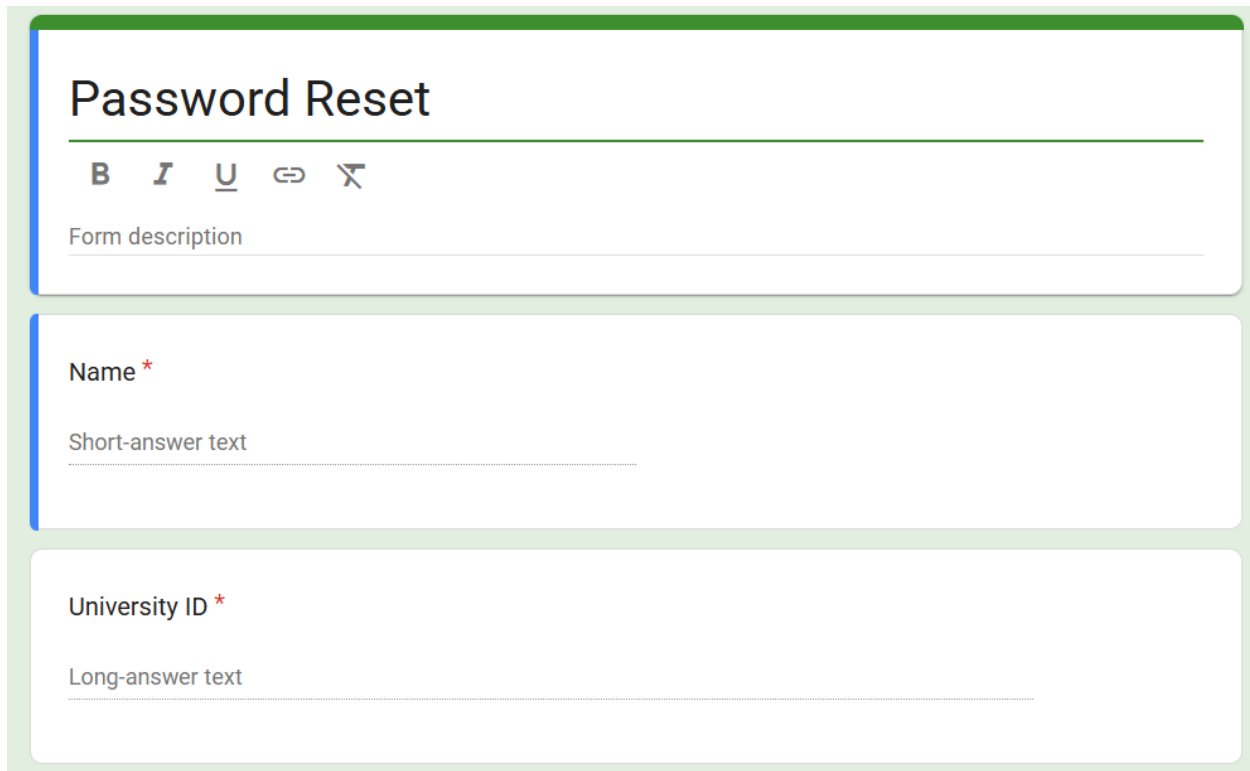
5.3 Psychological Techniques Used

Technique	How It Was Applied
Authority	Email appeared to come from the official IT department with a real sounding name, signature, and contact details.
Urgency	A hard deadline of March 3rd was set, with account suspension as the stated consequence of inaction.
Legitimacy Markers	A full professional signature block including helpdesk phone number and email address was included.
Deceptive Hyperlink	The visible link text showed a convincing domain while the actual URL pointed to a Google Form.
Fear	The threat of losing access to university systems motivated quick action without verification.
Reduced Suspicion	The 'disregard if already reset' line gave a sense of normalcy and reduced alarm.

5.4 Landing Page (Google Form)

The landing page was a Google Form designed to simulate a university password reset page. It requested the participant's Student ID and new password. Upon submission, no data was stored as passwords. Immediately after submitting, participants were shown an awareness banner explaining the simulation.

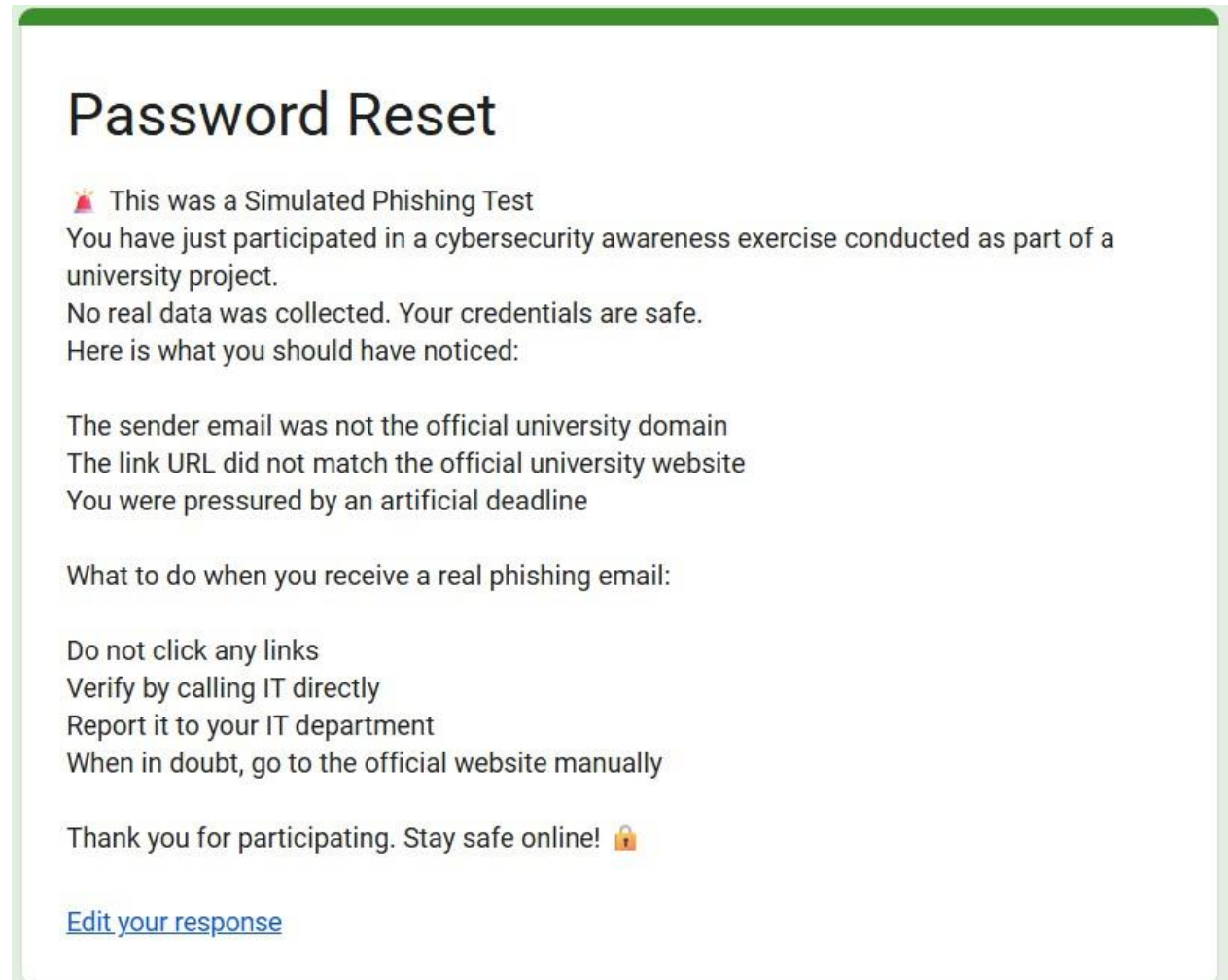
Screenshot 2: *Google Form landing page shown after clicking the link*



The screenshot shows a Google Form titled "Password Reset". The form has a green header bar. Below the title, there is a text area for "Form description" with formatting icons (B, I, U, link, unlink). The form contains two required text input fields. The first field is labeled "Name *" and has a "Short-answer text" input type. The second field is labeled "University ID *" and has a "Long-answer text" input type. The form is set against a light green background.

5.5 Awareness Banner

After form submission, participants were shown the following awareness message as the Google Form confirmation screen:



6. Results & Data Analysis

6.1 Campaign Results Summary

Metric	Count	Percentage
Total Emails Sent	6	100%
Forms Submitted (Fell for phishing)	4	66.7%
No Action Taken	2	33.3%
Reported as Suspicious	Unknown	—

6.2 Analysis

A success rate of 66.7% is significantly high for first-time phishing simulations in organizations without prior security awareness training. Studies indicate that untrained individuals fall for phishing attempts at rates between 25–80%, placing this result within the expected range.

The two participants who did not respond may have either identified the email as suspicious, ignored it, or simply did not open it. Without click tracking data, it is not possible to distinguish between these outcomes. This highlights a limitation of the current methodology and an area for improvement in future exercises.

Screenshot 3: *Google Sheets response log showing submission data*

5 responses

View in Sheets

Summary

Question

Individual

Name

5 responses

Rabya

hadiya zuberi

Haleema

Muhammad maaz

Jalisha

University ID

5 responses

245373

245359

245053

245011

245113

7. Red Flags Analysis

The phishing email contained several deliberate and unintentional red flags. These are documented below along with an assessment of whether participants were likely to notice them:

Red Flag	Why It Matters	Likelihood of Detection
Gmail sender address	Real IT departments never use Gmail	High — most overlooked this
Visible recipient list (CC)	Real mass emails use BCC	Medium — noticeable on close inspection
'IT_Department' with underscore	Unusual naming convention	Low — subtle formatting issue
Generic 'Dear Student'	Real emails use your name	Medium — commonly noticed in hindsight
Fake domain in link	airuniversity-portal vs airuniversity	High — requires careful reading
Artificial deadline	Pressure tactic — verify before acting	Low — urgency overrides suspicion

The post submission awareness banner explicitly highlighted each of these red flags to educate participants on what they should have looked for.

8. Preventive Measures

Based on the results of this exercise, the following preventive measures are recommended for individuals and institutions:

8.1 For Individual Users

- Always verify the sender's actual email address, not just the display name.
- Hover over any link before clicking to check the real destination URL.
- Be suspicious of any email that creates artificial urgency or threatens negative consequences.
- When in doubt, navigate to the official website manually rather than clicking email links.
- Contact IT support directly by phone to verify any password reset requests.
- Look for personalization, legitimate institutional emails use your name, not 'Dear Student'.

8.2 For Institutions

- Conduct regular phishing simulation exercises to maintain staff and student awareness.
- Display email security warnings for messages originating from outside the institution's domain.
- Establish a clear reporting channel for suspected phishing attempts.
- Include cybersecurity awareness as a mandatory component of student and staff onboarding.

9. Conclusion

This phishing awareness campaign successfully demonstrated that a well-crafted phishing email can deceive a significant proportion of university students even when observable red flags are present. A success rate of 66.7% underscores the critical need for ongoing security awareness education.

The exercise also highlighted the effectiveness of psychological manipulation techniques including urgency, authority impersonation, and deceptive hyperlinks. These are the same techniques used in real-world attacks and are particularly effective against individuals who have not received formal security awareness training.

Cybersecurity is ultimately a human problem as much as a technical one. Building a culture of vigilance and skepticism is the most effective long-term defense against phishing attacks.

10. Appendix

Appendix A: Full Phishing Email Text

Subject: Password Reset Required — Action Needed by March 3rd

Dear Student,

As part of our system upgrade to the new Student Portal, all students are required to verify and reset their account passwords.

You must complete this before March 3rd, 2026, or your account will be temporarily suspended.

Please reset your password using the secure link below:
<portal.airuniversity-portal.edu.pk/reset>

If you have already reset your password, please disregard this email.

Regards,
Ali Hassan
IT Support Team
Air University Aerospace and Aviation Campus, Kamra
helpdesk@airuniversitykamra.edu.pk | 051-517263

Appendix B: Raw Response Data

Screenshot 4: Full Google Sheets response log

Form_Responses		
Timestamp	Name	University ID
01/03/2026 13:48:52	Muhammad maaz	245011
01/03/2026 13:51:31	Jalisha	245113
01/03/2026 13:53:44	Haleema	245053
01/03/2026 13:58:54	Rabya	245373
01/03/2026 16:26:38	hadiya zuberi	245359